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Project Documentation

Operating System

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📄 SYSTEM MONITORING DASHBOARD

(Final Project Documentation)

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1. Introduction

In today's computer systems, monitoring performance is very important to ensure smooth and efficient working. Users often face problems like system slowdowns, high CPU usage, low memory, and disk space issues.

This project is a **Windows-based System Monitoring Dashboard** developed in **C++**. It provides real-time information about system performance and alerts the user when the system is under stress.

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2. Purpose of the Project

The main purpose of this project is:

- To monitor system performance in real-time
- To detect abnormal conditions (high CPU, RAM, disk usage)
- To provide alerts and suggestions to the user
- To create a visually interactive dashboard

In simple words:
This system not only monitors but also helps the user take action.

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3. Objectives

- Display CPU usage
- Display RAM usage
- Display Disk usage
- Show system uptime
- Provide voice alerts
- Show suggestions when system is overloaded
- Create a smooth and animated user interface

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4. Technologies Used

Technology	Purpose
C++	Programming language
Windows API	System-level functions
GDI+	Graphics and UI design
SAPI	Voice alerts (Text-to-Speech)

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5. How the System Works

The system works in the following steps:

Step 1: Data Collection

The system collects:

- CPU usage
- RAM usage
- Disk usage
- Uptime

Step 2: Data Processing

- Values are calculated and smoothed
- Compared with predefined limits

Step 3: Decision Making

- If usage is high → system becomes **CRITICAL**
- Otherwise → system is **OPTIMAL**

Step 4: Output

- Display values on screen
- Show status message
- Give suggestion
- Trigger voice alert

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6. System Features

Real-Time Monitoring

Updates system data every few milliseconds.

Animated Dashboard

- Moving particles
- Neon grid
- Cyber-style interface

Voice Alerts

Speaks warning messages when system is overloaded.

Smart Suggestions

Example:

- Close heavy apps
- Increase virtual memory
- Delete unnecessary files

Smooth UI

Uses smoothing factor to avoid sudden jumps in values.

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7. System Thresholds

Component	Limit
CPU	80%
RAM	75%
Disk	90%

When these limits are exceeded → system shows alert.

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8. Project Workflow

Step 1: Setup Environment

- Installed Visual Studio
- Configured Windows project

Step 2: Added Libraries

- GDI+ for graphics
- SAPI for voice

Step 3: Created Data Functions

- CPU usage function
- RAM usage function
- Disk usage function

Step 4: Designed UI

- Created dashboard layout
- Added animations

Step 5: Implemented Logic

- Threshold checking
- Status updates

Step 6: Added Alerts

- Sound + voice alerts

Step 7: Testing

- Ran program
- Checked alerts and UI

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9. Why This Project is Important

This project is useful because:

- Helps users understand system performance
- Prevents system crashes
- Improves system efficiency
- Provides real-time feedback
- Enhances user experience

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10. Advantages

- Easy to use

- Real-time monitoring
- Interactive UI
- Voice-based alerts
- Smart suggestions

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11. Conclusion

This project successfully demonstrates a **real-time system monitoring dashboard** using C++. It not only displays system performance but also provides alerts and suggestions when needed.

Final statement:

“This system improves system awareness by combining monitoring, visualization, and intelligent alerting in one application.”